Can LIVI latent factors be used as quantitative traits for genetic association testing?

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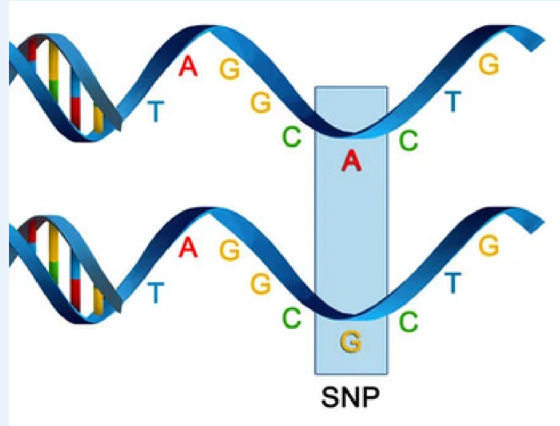
Biological Background

SNPs are common genetic variants. They can together increase an individual's genetic risk for diseases. Most SNPs are found on non-coding regions of the DNA.



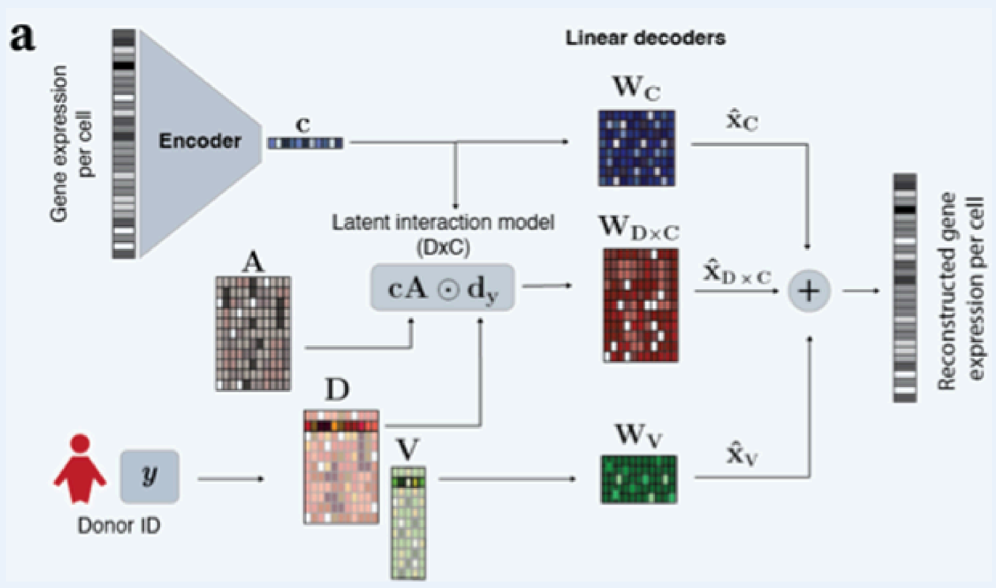
Gene expression is how much genes certain genes are used. Despite SNPs mainly lying outside of genes, they still influence gene expression.

Previous work has directly tested between SNPs and gene expression. This has the following problems:

- There are a lot of SNPs and a lot of genes. Testing between all combinations is both computationally expensive and subject to the multiple testing problem, where false positives occur by chanche.
- Effects of SNPs are often small and get diluted when tested individually. Current research mainly finds SNPs that act on nearby genes

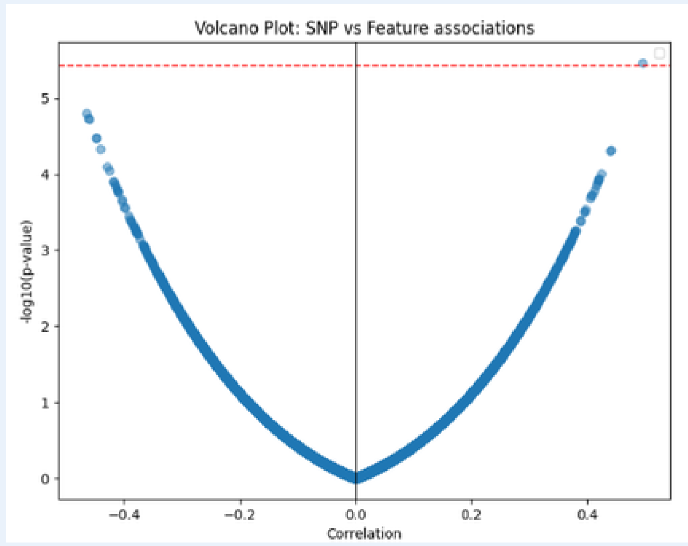
Computer Science Background

LIVI is a recently develloped variational auto-encoder (VAE). The latent space of a VAE is meaningfully structured. LIVI operates on single-cell gene expression data and encodes this into seperate latent spaces for Donor and Cell effects.

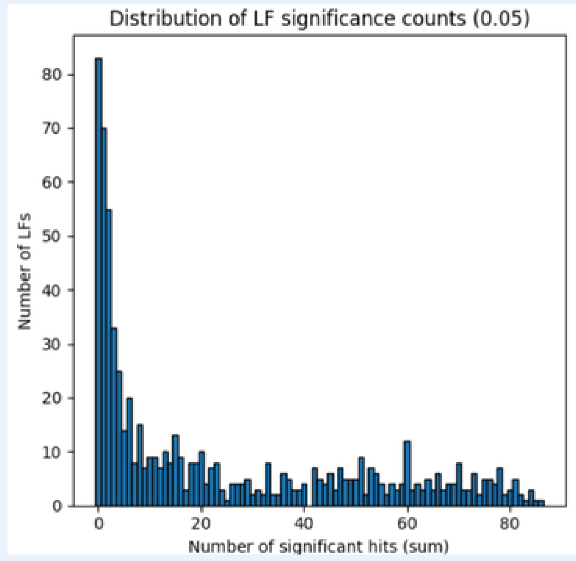


Results

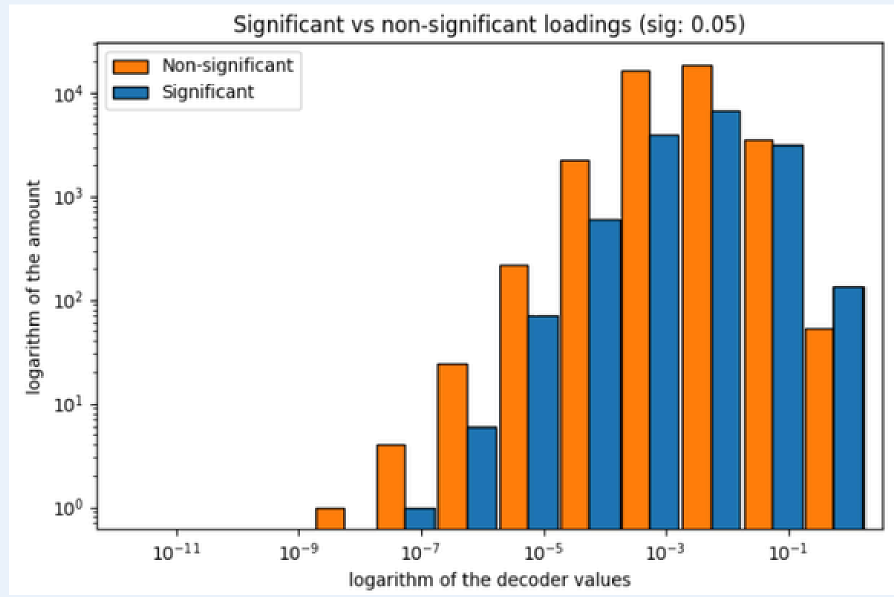
A **direct linear correlation is insuficient.** Pearsons correlation between all 65100 SNP latent factor pairs resulted in only one siginificant result. This is due to it not taking covariates and other confounders into account



A Linear Mixed Model substantially increases results. By separating random and fixed effects, the Linear Mixed Model found 16650 significant associations across 617 out of 700 latent factors in the D-embedding.



GWAS identified SNP-gene pairs show higher decoding effect for significant associations. The loadings of the pairs from the GWAS show higher loading in the decoding matrix DxC. Functionally mapped GWAS genes ranked significantly higher in the decoders of significant associations than non-significant ones.



Methods

Data used

- We applied and trained LIVI on a dataset containing data from 314.000 cells from 72 Rheumatoid Arthritis patients.
- The SNPs we used are previously identified to affect risk of RA via a Genome Wide Association Study (GWAS), a large scale study amongst different ancestries.

LIVI

- We used 700 D factors and 5 V factors for each individual, and 15 C factors for each single-cell-measurement.
- For the association testing the D-embedding was used, since the DNA and thus the SNPs are the same across all cells

Linear Mixed Model

$$d_k = g\beta + M\alpha + u + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2 I)$$

$$\text{where} \quad u \sim \mathcal{N}(0, \sigma_u^2 K)$$

- g is a vector of SNP genotypes
- d_k are the latent variables
- β is the fixed effect size to be estimated
- M is the matrix containing confounders
- K is the Kinship matrix containg population structure